

## Practice Problems 4

ECON 441: Introduction to Mathematical Economics

Instructor: Div Bhagia

### Exercise 4.4

5. (e) Find (i)  $C = AB$ , and (ii)  $D = BA$ , if

$$A = \begin{bmatrix} -2 \\ 4 \\ 7 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 6 & -2 \end{bmatrix}$$

7. If the matrix  $A$  in Example 5 had all its four elements nonzero, would  $x'Ax$  still give a weighted sum of squares? Would the associative law still apply?

### Exercise 4.5

1. Given  $A = \begin{bmatrix} -1 & 5 & 7 \\ 0 & -2 & 4 \end{bmatrix}$ ,  $b = \begin{bmatrix} 9 \\ 6 \\ 0 \end{bmatrix}$ , and  $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$  :

Calculate: (a)  $AI$  (b)  $IA$  (c)  $Ix$  (d)  $x'I$

Indicate the dimension of the identity matrix used in each case.

4. Show that the diagonal matrix

$$\begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}$$

can be idempotent only if each diagonal element is either 1 or 0 . How many different numerical idempotent diagonal matrices of dimension  $n \times n$  can be constructed altogether from such a matrix?

### Exercise 5.1

3. Are the rows linearly independent in each of the following?

a)  $\begin{bmatrix} 24 & 8 \\ 9 & -3 \end{bmatrix}$

b)  $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

c)  $\begin{bmatrix} 0 & 4 \\ 3 & 2 \end{bmatrix}$

d)  $\begin{bmatrix} -1 & 5 \\ 2 & -10 \end{bmatrix}$

4. Check whether the columns of each matrix in Prob. 3 are also linearly independent. Do you get the same answer as for row independence?